



Loop

dot
LEBANON

Activate Windows
Go to Settings to activate Windows

© 2020 Microsoft Corporation. All rights reserved. Microsoft, the Microsoft Dynamics logo, and the Microsoft Dynamics logo are either registered trademarks or trademarks of Microsoft Corporation in the United States and/or other countries.

Free trial

100%

Objectives

Define the Loop

Use the "for" and the "while" loop

Differentiate between "for" and "while" loop

Use the while loop to count the time

5- Loop - PowerPoint (Product Activation Failed)

File Home Insert Design Transitions Animations Slide Show Review View Developer Acrobat Tell me what you want to do... Sign in Share

Cut Copy Paste Format Painter New Slide Section Layout Reset

Font Paragraph Drawing Editing

Shape Fill Shape Outline Shape Effects Find Replace Select Create a PDF Adobe Acrobat

1

Loop

A loop is a programming structure that repeats a sequence of instructions until a specific condition is met.

- It saves time and prevents redundancy, allowing developers to execute tasks like processing lists or running calculations thousands of times without writing the same code repeatedly.
- The Loop is a special statement that keeps repeating until the conditions are met (true).
- We can see the loop in each program, at the void loop, which will be repeating the main body of the program.

dot

1

Slide 4 of 27 Arabic (Lebanon)

Search the web and Windows

mscrt.google.com is sharing your screen. Stop sharing Hide

Notes Comments

3:07 PM 7/2026

5- Loop - PowerPoint (Product Activation Failed)

File Home Insert Design Transitions Animations Slide Show Review View Developer Acrobat Tell me what you want to do... Sign in Share

Paste Cut Copy Format Painter Clipboard Layout Reset Section New Slide Slides Font Paragraph Text Direction Align Text Convert to SmartArt Drawing Shape Fill Shape Outline Shape Effects Find Replace Select Create a PDF Adobe Acrobat

1 2 3 4 5 6

“For” and “While” loop

Activate Windows Go to Settings to activate Windows.

meet.google.com is sharing your screen. Stop sharing Hide

Slide 5 of 27 C.A. Arabic (Lebanon)

Notes Comments

5- Loop - PowerPoint (Product Activation Failed)

File Home Insert Design Transitions Animations Slide Show Review View Developer Acrobat Tell me what you want to do... Sign in Share

Cut Copy Paste Format Painter Clipboard Layout Reset New Slide Section Slides Font Paragraph Drawing Shape Fill Shape Outline Shape Effects Find Replace Select Create a PDF Adobe Acrobat

"For" and "While" loop

The loop can help in counting, repeat a certain task, or test for a repeatable number.

- The **"for loop"** is used when counting (predetermined number of times).
- The **"while loop"** is used based on sensors readings or time (runs indefinitely until a specific sensor condition or event changes).

For Loop

```
graph TD; Start([Start]) --> LastItem{Last Item?}; LastItem -- No --> Statements[/Statements/]; LastItem -- Yes --> End([End]); Statements --> End;
```

While Loop

```
graph TD; Start([Start]) --> Condition{Condition}; Condition -- True --> Statements[/Statements/]; Statements --> Condition; Condition -- False --> End([End]);
```

dot: [dot.org](#)

Activate Windows Go to Settings to activate Windows.

meet.google.com is sharing your screen. Stop sharing Hide

Notes Comments 116R

Search the web and Windows

3:22 PM 7/1/2026

"For" and "While" loop

The loop can help in counting, repeat a certain task, or test for a repeatable number.

- The "for loop" is used when counting (predetermined number of times).
- The "while loop" is used based on sensors readings or time (runs indefinitely until a specific sensor condition or event changes).

For Loop



While Loop



"For" loop

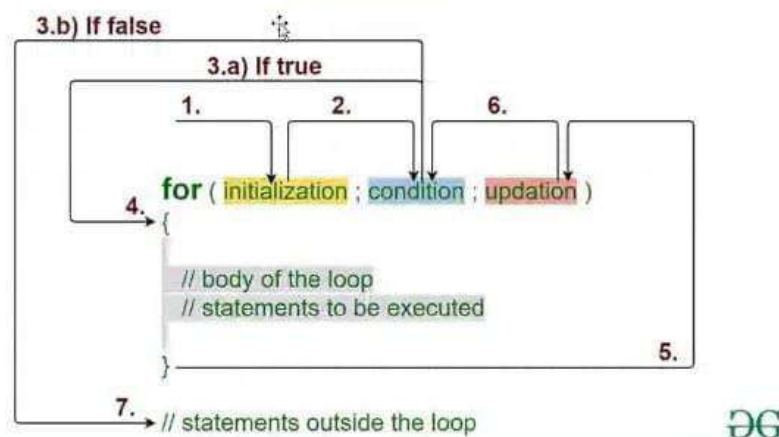
For (①variable defining, ②condition, ④action)

{③Loop action};

1. This part will be executed for one time. It is mainly used for variable defining.
2. The second step is testing the condition. If the condition is true, it will enter the loop.
3. The third stage is to execute the loop actions
4. The fourth stage is the loop ending action, which will be executed only once after each loop

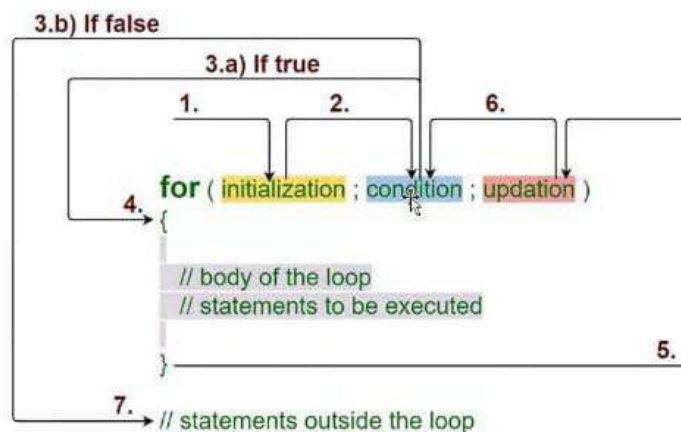
"For" loop

For Loop



"For" loop

For Loop



meet.google.com is sharing your screen.

Stop sharing

Hide

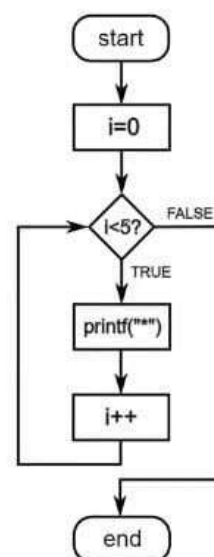
Activate Windows
Go to Settings to activate Windows.

"For" loop

Example

```
For (int i = 0 ; i < 5 ; i++ ) {  
  digitalWrite( LED , High);  
  delay(1000);  
  digitalWrite( LED, Low);  
  delay(1000);  
}
```

This loop will blink the LED 5 times.



"For" loop

```
for ( int i = 5 ; i <= 10 ; i + + )
```

The diagram illustrates the three components of a C++ for loop. The code `for ( int i = 5 ; i <= 10 ; i + + )` is shown with brackets underneath each part. Arrows point from these brackets to the labels: `int i = 5` is labeled "Initialization", `i <= 10` is labeled "Test Condition", and `i + +` is labeled "Updation".

C++ for Loop



"For" loop

```
for ( int i = 5 ; i <= 10 ; i ++ )
```

The diagram illustrates the three components of a C++ for loop. The code `for ( int i = 5 ; i <= 10 ; i ++ )` is shown with brackets and arrows pointing to labels below. The first part, `int i = 5`, is labeled "Initialization". The second part, `i <= 10`, is labeled "Test Condition". The third part, `i ++`, is labeled "Updation".

C++ for Loop

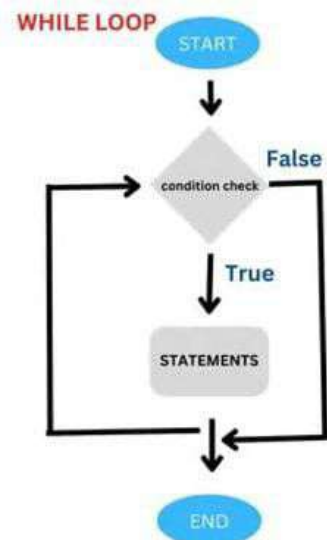


"While" loop

The "while loop", will only test the condition then execute the code.

While (①Condition) { ②loop Action }

1. First, the loop will test for the condition.
2. If the loop, the condition is correct then the loop actions will start.



"While" loop

Example

```
While ( i<5){  
  digitalWrite( LED , High);  
  delay(1000);  
  digitalWrite( LED, Low);  
  delay(1000);  
  i++;  
}
```

This loop will blink the LED 5 times.

But you need to define the variable `i` before the loop.

```
int count = 1;  
while(count <= 10)  
{  
  System.out.println(count);  
  count++;  
}
```

1. Initialize loop variable

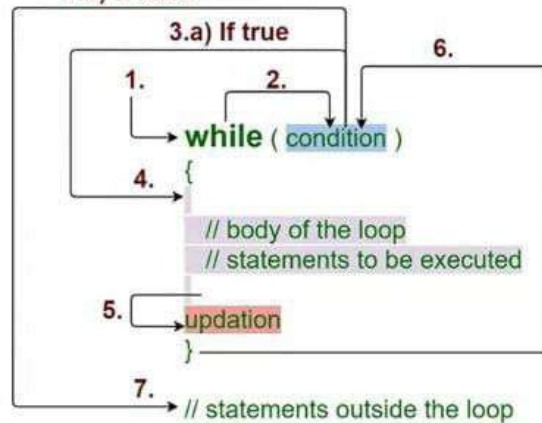
2. Test loop variable

3. Change loop variable

"While" loop

While Loop

3.b) If false



dot.

Activate Windows
Go to Settings to activate Windows.

meet.google.com is sharing your screen.

Stop sharing

Hide

Notes Comments

Const int

- The `int` is basically the type of integer type data and `const` is used to make something constant.
- A `const int` is a read-only integer. Once it is initialized, its value cannot be changed for the duration of the program allowing the compiler to optimize and completely eliminate RAM usage.
- In contrast, a standard `int` is a mutable integer. Its value can be reassigned, modified, and updated as the program runs.

```
int score = 10;           // A standard int (can be changed)
const int maxScore = 50; // A const int (cannot be changed)

score = 20;               // Allowed: score is now 20
maxScore = 60;            // ERROR: Attempting to modify a read-only variable
```

dot
LEARNON

meet.google.com is sharing your screen.

Stop sharing

Hide

Notes

Comments

Activate Windows
Go to Settings to activate Windows.

5- Loop - PowerPoint (Product Activation Failed)

File Home Insert Design Transitions Animations Slide Show Review View Developer Acrobat Tell me what you want to do... Sign in Share

Clipboard Font Paragraph Drawing Editing Adobe Acrobat

Const int

- The int is basically the type of integer type data and const is used to make something constant.
- A const int is a read-only integer. Once it is initialized, its value cannot be changed for the duration of the program allowing the compiler to optimize and completely eliminate RAM usage.
- In contrast, a standard int is a mutable integer. Its value can be reassigned, modified, and updated as the program runs.

```
int score = 10;           // A standard int (can be changed)
const int maxScore = 50; // A const int (cannot be changed)

score = 20;               // Allowed: score is now 20
maxScore = 60;            // ERROR: Attempting to modify a read-only variable
```

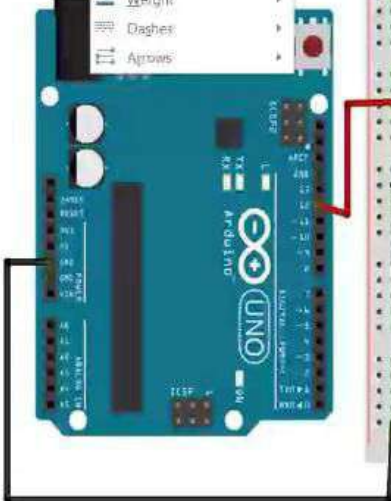
dot

Activate Windows Go to Settings to activate Windows.

meet.google.com is sharing your screen. Stop sharing Hide

Notes Comments 116% 3:57 PM ENG

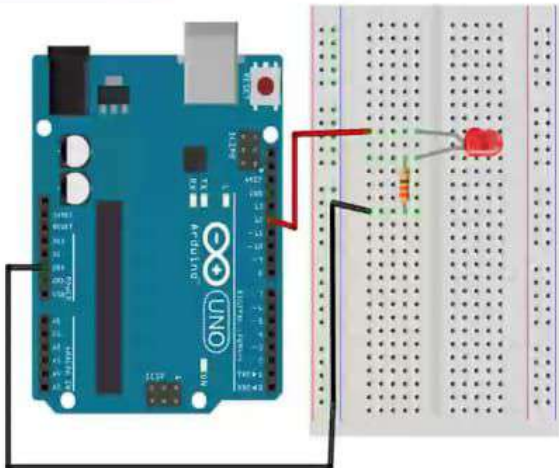
Board connection





Animations Slide Show Review View Developer Aprobat Tell me what you want to do

Board connection



4

meet.google.com is sharing your screen. Stop sharing Hide

Stop sharing

 Notes
 Comments

Mohamad is presenting



21 others

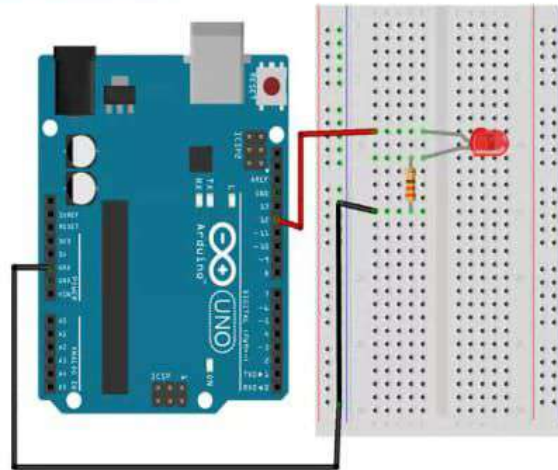




5- Loop - PowerPoint (Product Activation Failed)



Board connection



dot

meet.google.com is sharing your screen.

Stop sharing

Hide

Notes Comments



Mohamad is presenting



You

21 others



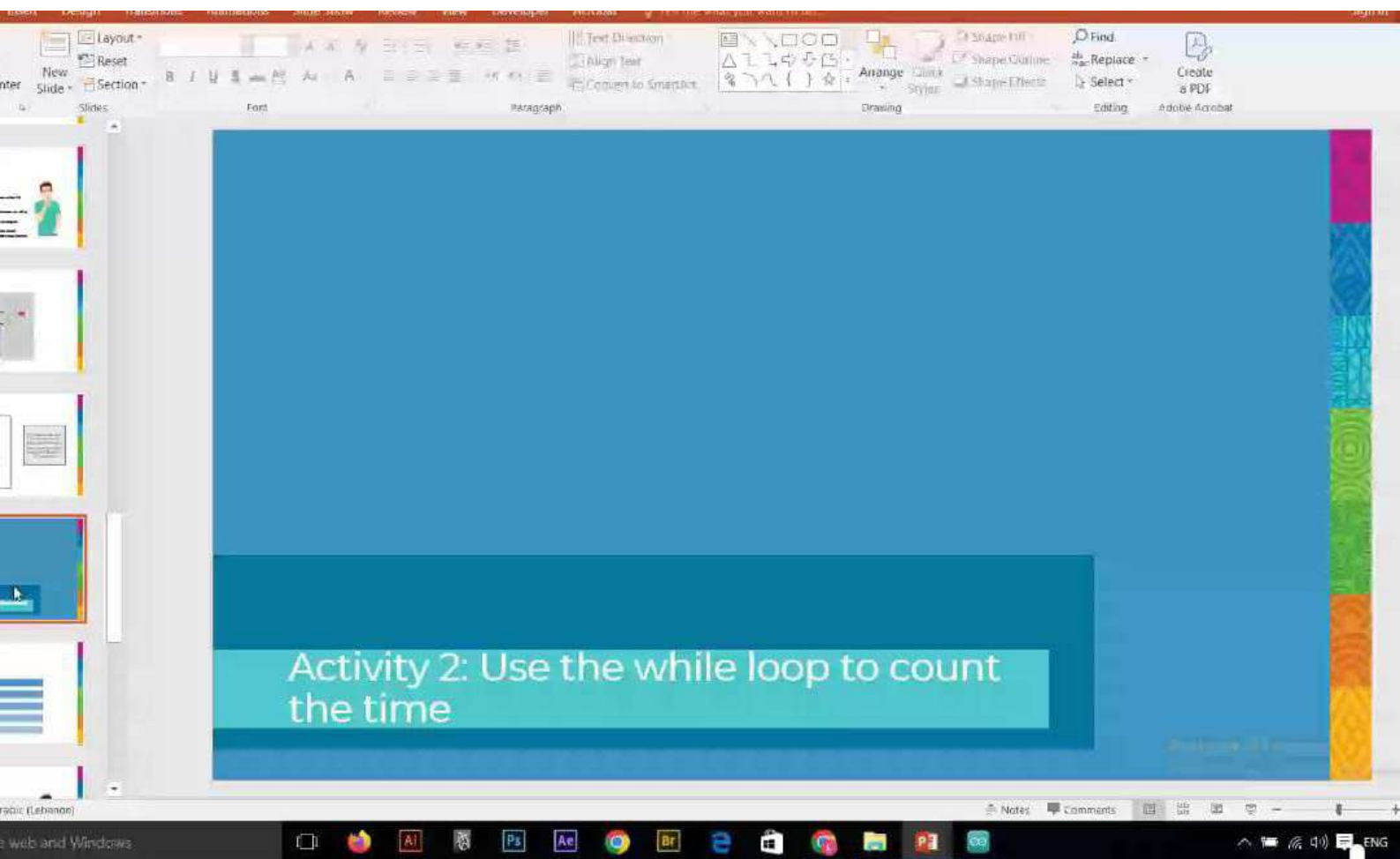
Mohamad

```
1 int led=9;
2
3 void setup() {
4   // put your setup code here, to run once:
5   pinMode(led,OUTPUT);
6 }
7
8 void loop() {
9   // put your main code here, to run repeatedly:
10  for(int i=0;i<5;i++){ //for(int i=1; i<=5; i++) or for(int i =1; i<6;i++)
11    digitalWrite(led,HIGH);
12    delay(1000);
13    digitalWrite(led,LOW);
14    delay(1000);
15  }
16  delay(5000); //delay of 5 seconds to differentiate the execution of the loop counting 5 times
17  //exit(1); //this instruction wont repeat the void loop again
18 }
19
```



```
1 int led=9;
2
3 void setup() {
4   // put your setup code here, to run once:
5   pinMode(led,OUTPUT);
6 }
7
8 void loop() {
9   // put your main code here, to run repeatedly:
10  for(int i=0;i<5;i++){ //for(int i=1; i<=5; i++) or for(int i =1; i<6;i++)
11    digitalWrite(led,HIGH);
12    delay(1000);
13    digitalWrite(led,LOW);
14    delay(1000);
15  }
16  //delay(5000); //delay of 5 seconds to differentiate the execution of the loop counting 5 times
17  exit(1); //this instruction wont repeat the void loop again
18 }
19
```

```
1 int led=9;
2
3 void setup() {
4   // put your setup code here, to run once:
5   pinMode(led,OUTPUT);
6 }
7
8 void loop() {
9   // put your main code here, to run repeatedly:
10  for(int i=0;i<5;i++){ //for(int i=1; i<=5; i++) or for(int i =1; i<6;i++)
11    digitalWrite(led,HIGH);
12    delay(1000);
13    digitalWrite(led,LOW);
14    delay(1000);
15  }//the code will blink the led 5 times
16  //delay(5000);//delay of 5 seconds to differentiate the execution of the loop counting 5 times
17  exit(1); //this instruction wont repeat the void loop again
18 }
19
```



5- Loop - PowerPoint (Product Activation failed)

Design Transitions Animations Slide Show Review View Developer Acrobat Tell me what you want to do... Sign in Share

Layout * Reset Section * Font Paragraph Drawing Editing Adobe Acrobat

Use the while loop to count the time

Required Elements:

- Arduino
- X2 Resistor 330 Ω
- Red LED
- Push Button
- Breadboard

dot

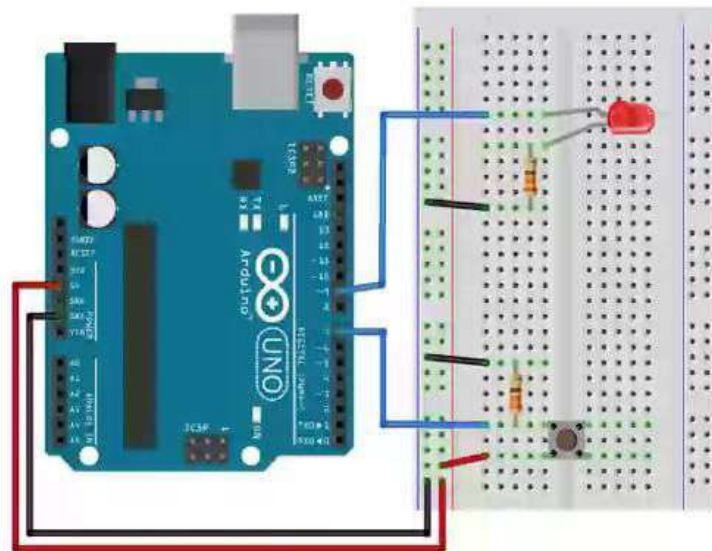
Activate Windows
Go to Settings to activate Windows.

Windows

Notes Comments

4:00 PM
7/20/2026

Board connection

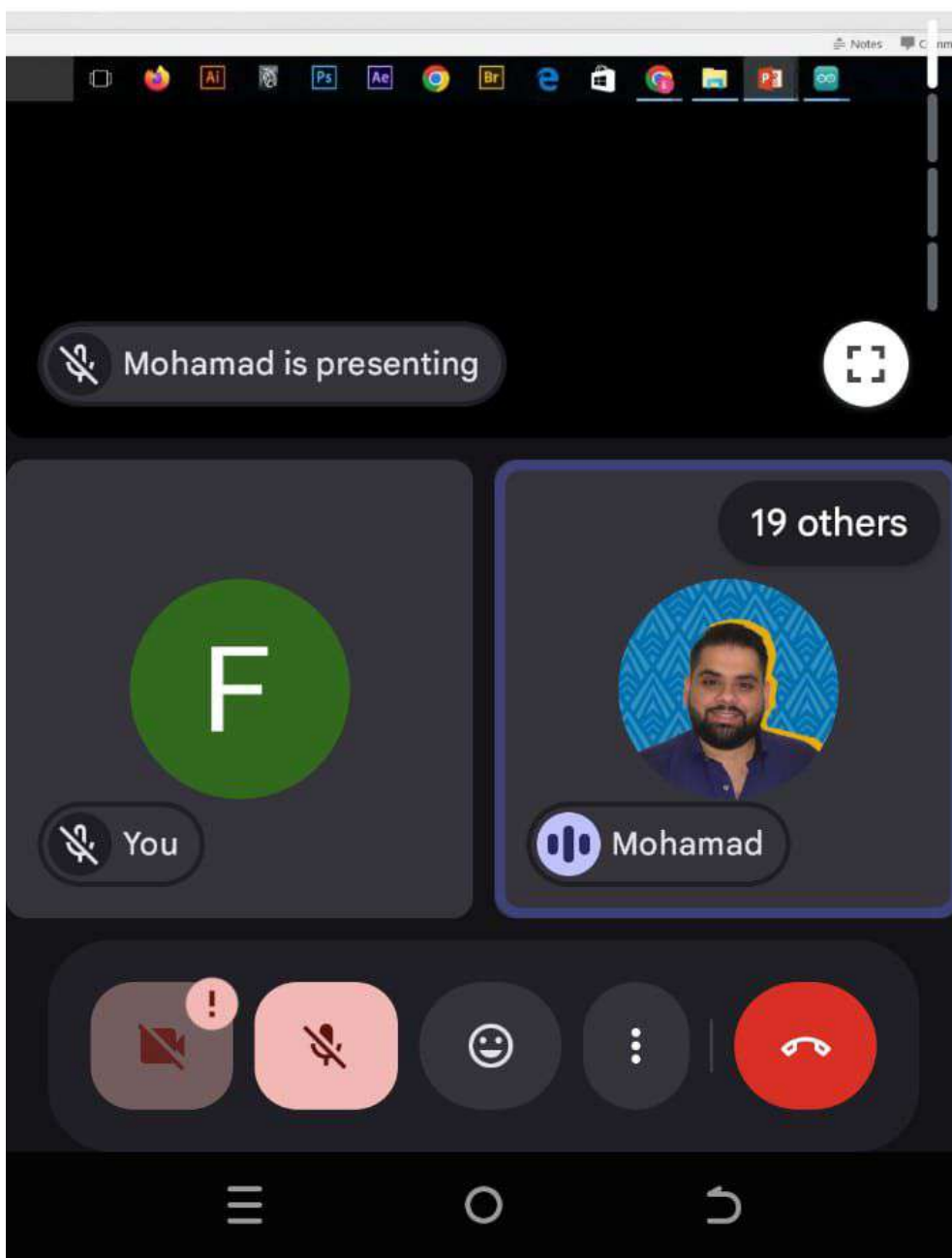
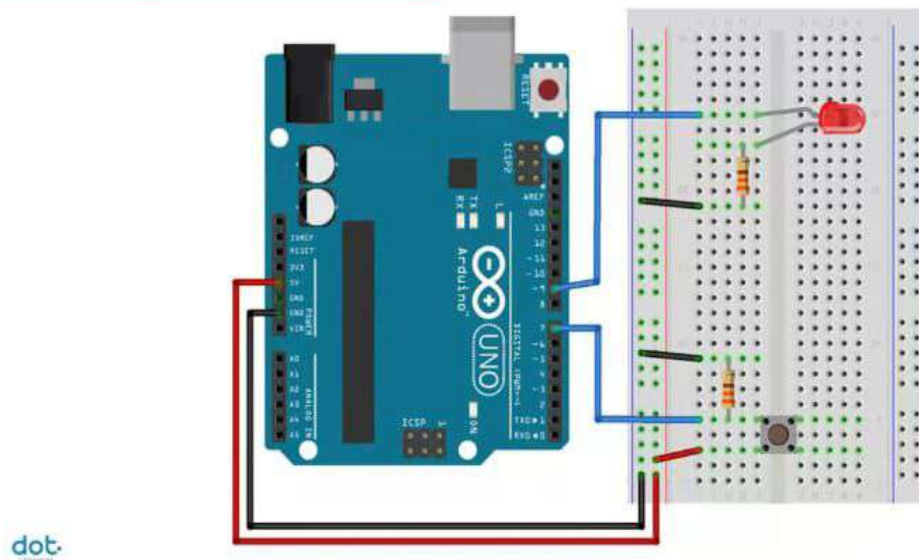


dot

Activate Windows
Go to Settings to activate Windows.



Board connection



File Home Insert Design Transitions Animations Slide Show Review View Developer Acrobat Tell me what you want to do... Sign in Share

Clipboard Cut Copy Paste Format Painter New Slide Section Layout Reset Slides Font Paragraph Drawing Shape Fill Shape Outline Shape Effects Find Replace Select Create a PDF Adobe Acrobat

Board connection

The diagram illustrates the connection of an Arduino Uno microcontroller to a breadboard-based circuit. The breadboard contains a blue integrated circuit (IC), likely a motor driver, and a small component, possibly a diode or a small motor. Wires connect the Arduino pins to the breadboard components. The connections are as follows:

- Red wire: Connects the 5V pin of the Arduino to the positive rail of the breadboard.
- Black wire: Connects the GND pin of the Arduino to the negative rail of the breadboard.
- Blue wire: Connects the digital output pin 12 of the Arduino to the input pin of the blue IC.
- Black wire: Connects the GND pin of the Arduino to the input pin of the blue IC.

dot: logo

Activate Windows
Go to Settings to activate Windows.

Slide 24 of 27 | Arabic (Lebanon) | Notes | Comments | 116%

5- Loop - PowerPoint (Product Activation failed)

File Home Insert Design Transitions Animations Slide Show Review View Developer Acrobat Tell me what you want to do... Sign in Share

Clipboard Font Paragraph Drawing Editing Adobe Acrobat

Use the while loop to count the time

How to think before Connecting

1. Start by defining five variables.
 - LED, Button, buttonState, secondCount, and i.
2. Define the pin type for the LED and the push button.
 - We can notice that the LED pin is an output, this is because we will be sending a signal to turn on the LED.
 - The second is the "button" which will have the push button pin number.
3. In the void loop function; this means that we are starting our program.
 - Use digitalRead to store the value from the button pin in "buttonState" variable.
 - Store zero value in the variables "secondCount", and "i".
 - Add a "While loop" and write the main conditions in it. the loop will only enter if it reads a HIGH voltage at the button PIN.



dot: logo

Activate Windows
Go to Settings to activate Windows.

Slide 22 of 27 Aratic (Lebanon)

Notes Comments



Rawan +23



```
sketch_jul20f | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
sketch_jul20f.ino
1 const int led=9;
2 const int button=7;
3 int ButtonState; // variable to give the button st
4 int SecondCount; //variable to count the time spe
5 int i; //loop variable
6
7 void setup() {
8   // put your setup code here, to run once:
9   pinMode(led,OUTPUT);
10  pinMode(button,INPUT);
11 }
12
13 void loop() {
14   // put your main code here, to run repeatedly
15   ButtonState= digitalRead(button);
16   i=0;
17   SecondCount=0;
18
19   while(ButtonState)
20   }
21
```

use we will be
n number.
program.
in "buttonState"
p will only enter

Activate Windows
Go to Settings to activate Windows.



Mohamad is presenting



You



21 others



Mohamad





Rawan +22



sketch_jul20f | Arduino IDE 2.3.10

File Edit Sketch Tools Help

Arduino UNO

```
1  const int button=7;
2
3  int ButtonState; // variable to give the button s
4  int SecondCount; //variable to count the time spe
5  int i; //loop variable
6
7  void setup() {
8      // put your setup code here, to run once:
9      pinMode(led,OUTPUT);
10     pinMode(button,INPUT);
11 }
12
13 void loop() {
14     // put your main code here, to run repeatedly:
15     ButtonState= digitalRead(button);
16     i=0;
17     SecondCount=0;
18     delay(unsigned long ms) ->
19     while(Butt void
20     Delay(100)
21 }
22 }
23
```

Activate Windows
Go to Settings to activate Windows

e variable "i" and

t one using the

ent pushing the

e we have spent

p". This will allow



Mohamad is presenting



You



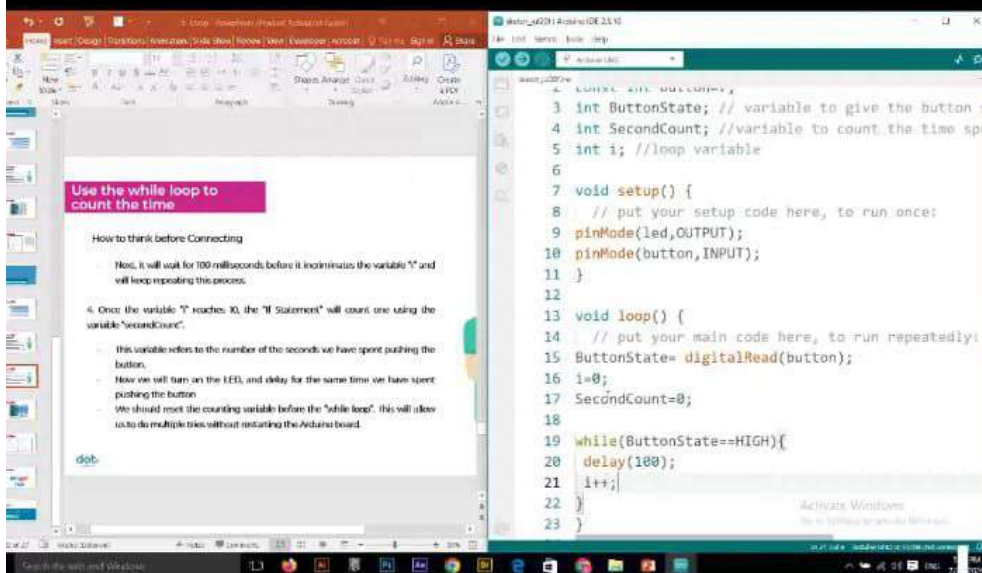
Mohamad

20 others





Rawan +22



Mohamad is presenting



You



Mohamad

20 others





Rawan +22



f | Arduino IDE 2.3.10

File Edit Sketch Tools Help

Arduino UNO

```
sketch_jul20f.ino
2  const int button=7;
3  int ButtonState; // variable to give the button st
4  int SecondCount; //variable to count the time spe
5  int i; //loop variable
6
7  void setup() {
8      // put your setup code here, to run once:
9      pinMode(led,OUTPUT);
10     pinMode(button,INPUT);
11 }
12
13 void loop() {
14     // put your main code here, to run repeatedly:
15     ButtonState= digitalRead(button);
16     i=0;
17     SecondCount=0;
18
19     while(ButtonState==HIGH){
20         delay(100);
21         i++;
22     }
23 }
```

Activate Windows

Go to Settings to activate Windows.



Mohamad is presenting

Ln 21, Col 5 Arduino UNO on COM



3 PM 7/20/2026



You

20 others



Mohamad





sketch_ju20f | Arduino IDE 2.3.10

File Edit Sketch Tools Help

sketch_ju20f.ino

```
14 // put your main code here, to run repeatedly:
15 ButtonState= digitalRead(button);
16 i=0;
17 SecondCount=0;
18
19 while(ButtonState==HIGH){
20   delay(100);
21   i++;
22   if(i==10){
23     SecondCount++; //command to count the time
24     i=0;
25   }
26   ButtonState=digitalRead(button);
27 }
28 if(SecondCount>0){
29   digitalWrite(led,HIGH);
30   delay(SecondCount*1000); // ton convert the time to ms
31 }
32 digitalWrite(led,LOW);
33 }
```

Output

Search the web and Windows



Mohamad is presenting



You



Mohamad

20 others

5:32 PM

14.3 K/S 4G 11



Rawan +22



```
12
13 void loop() {
14   // put your main code here, to run repeatedly:
15   ButtonState= digitalRead(button);
16   i=0;
17   SecondCount=0;
18
19   while(ButtonState==HIGH){
20     delay(100);
21     i++;
22     if(i==10){
23       SecondCount++; //command to count the time
24       i=0;
25     }
26     ButtonState=digitalRead(button);
27   }
28   if(SecondCount>0){
29     digitalWrite(led,HIGH);
30     delay(SecondCount*1000); // ton convert the time to ms
31   }
32   digitalWrite(led,LOW);
```

Activate Windows



Mohamad is presenting



20 others



Ahmad Youssef left



